

# In-class review

STAT 870 - Fall 2026

2026-09-09

1. Answer in one sentence: what is the experimental unit?

2. What are the assumptions behind this model?

$$y_i \sim N(\mu_i, \sigma^2)$$

3. What could happen if we have more than one observation per experimental unit and don't specify this in the model?

4. Complete the table below.

Table 1: Fixed vs Random Effects

|                                                                               | Fixed effects | Random effects |
|-------------------------------------------------------------------------------|---------------|----------------|
| Assumption about this effect                                                  |               |                |
| Affects the standard error of the marginal means (yes/no)                     |               |                |
| What parameter do we estimate?                                                |               |                |
| Usually used to model (e.g. treatments, elements of design (any exceptions?)) |               |                |
| Method of estimation                                                          |               |                |